Mitigating Motion Artifacts to Improve ECG Accuracy

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May 2025

Outline

1. Introduction
2. Project Goals
3. Methodology
4. Circuit Design and Assembly
5. Signal Processing
6. Results
7. Conclusion and Future Works
8. References

ECG Signals

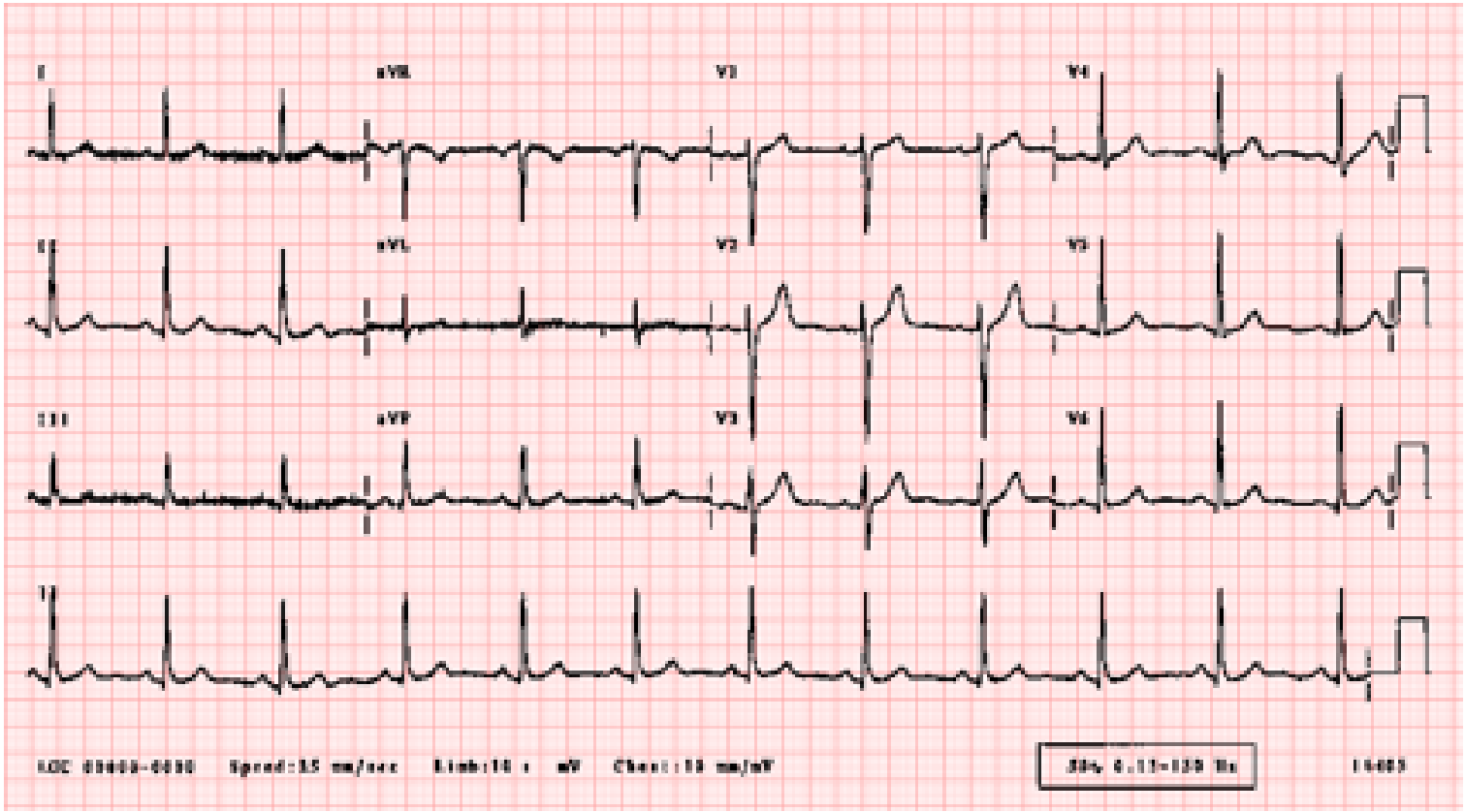


Figure: Clean ECG recording across multiple leads

ECG Signals

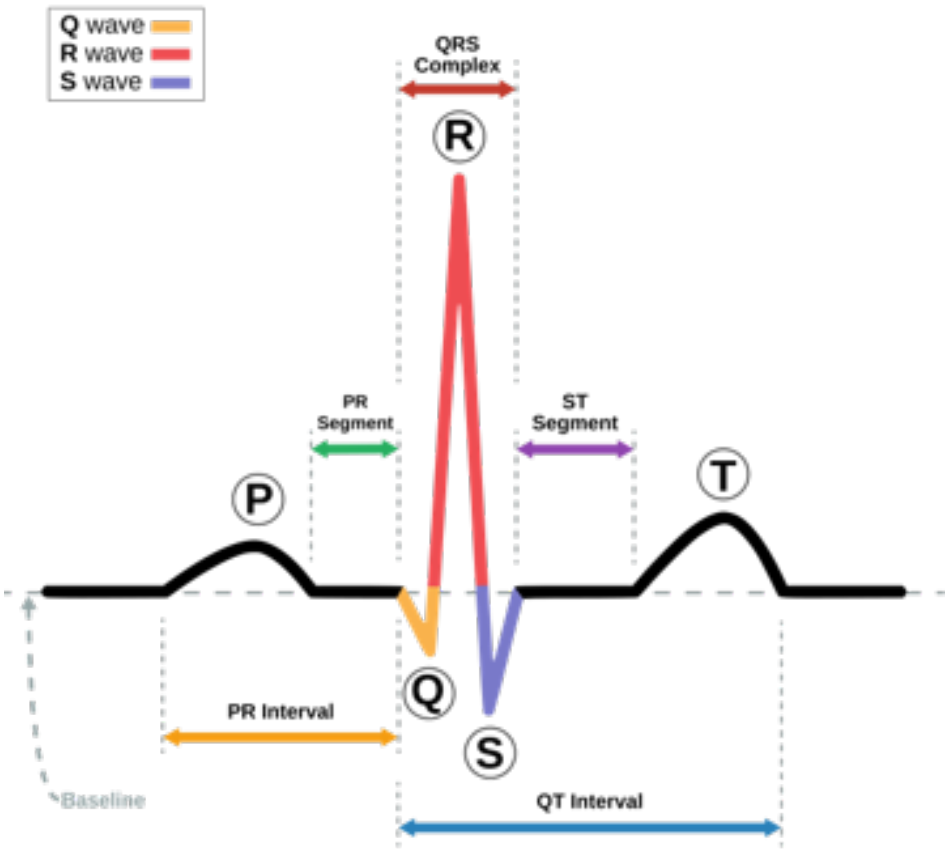


Figure: Annotated representation of the ECG waveform components

Common Motion Artifacts in ECG

- Breathing
- Physical activities
- Electrode shifts
- Muscle contractions
- External electrical interference

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Project Goal: Recognition of Motion Artifacts

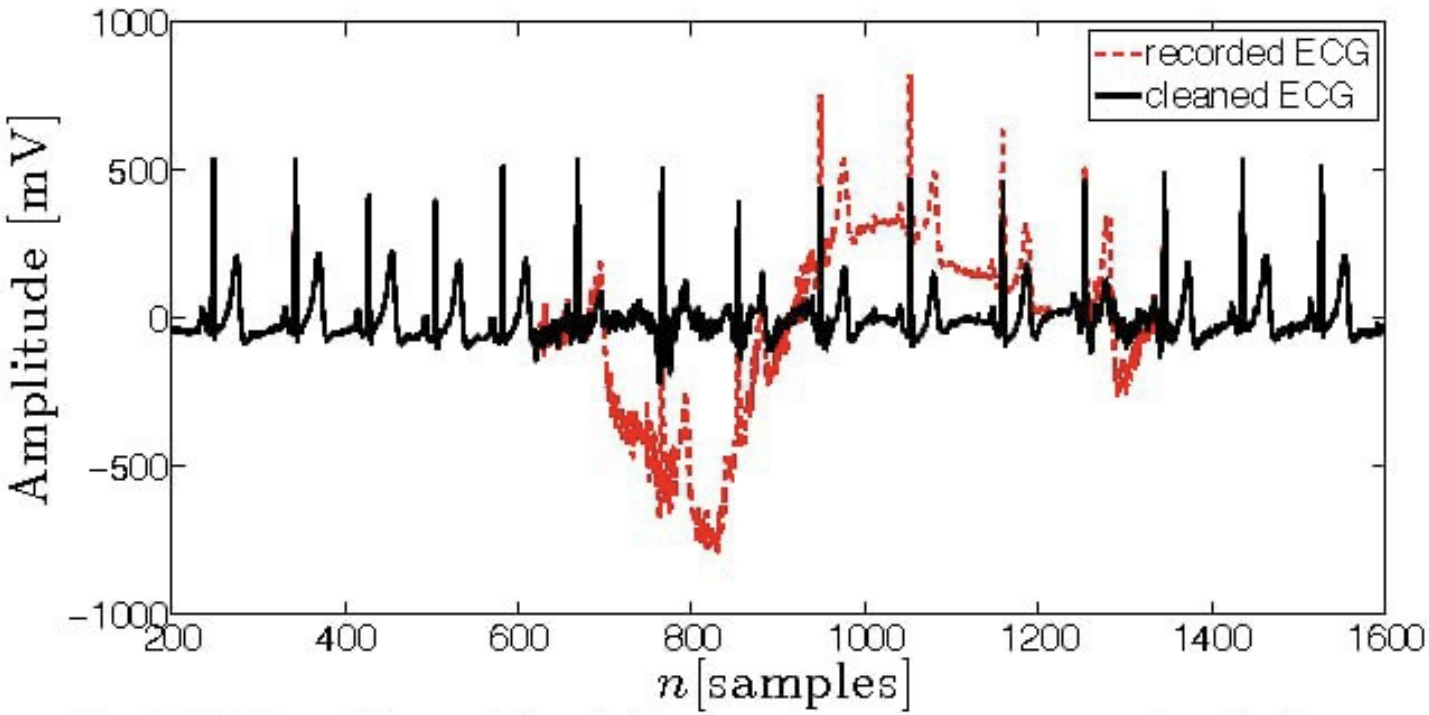


Figure: Comparison of Recorded vs. Cleaned ECG Signal

Project Goal: ECG Recording During Physical Activity

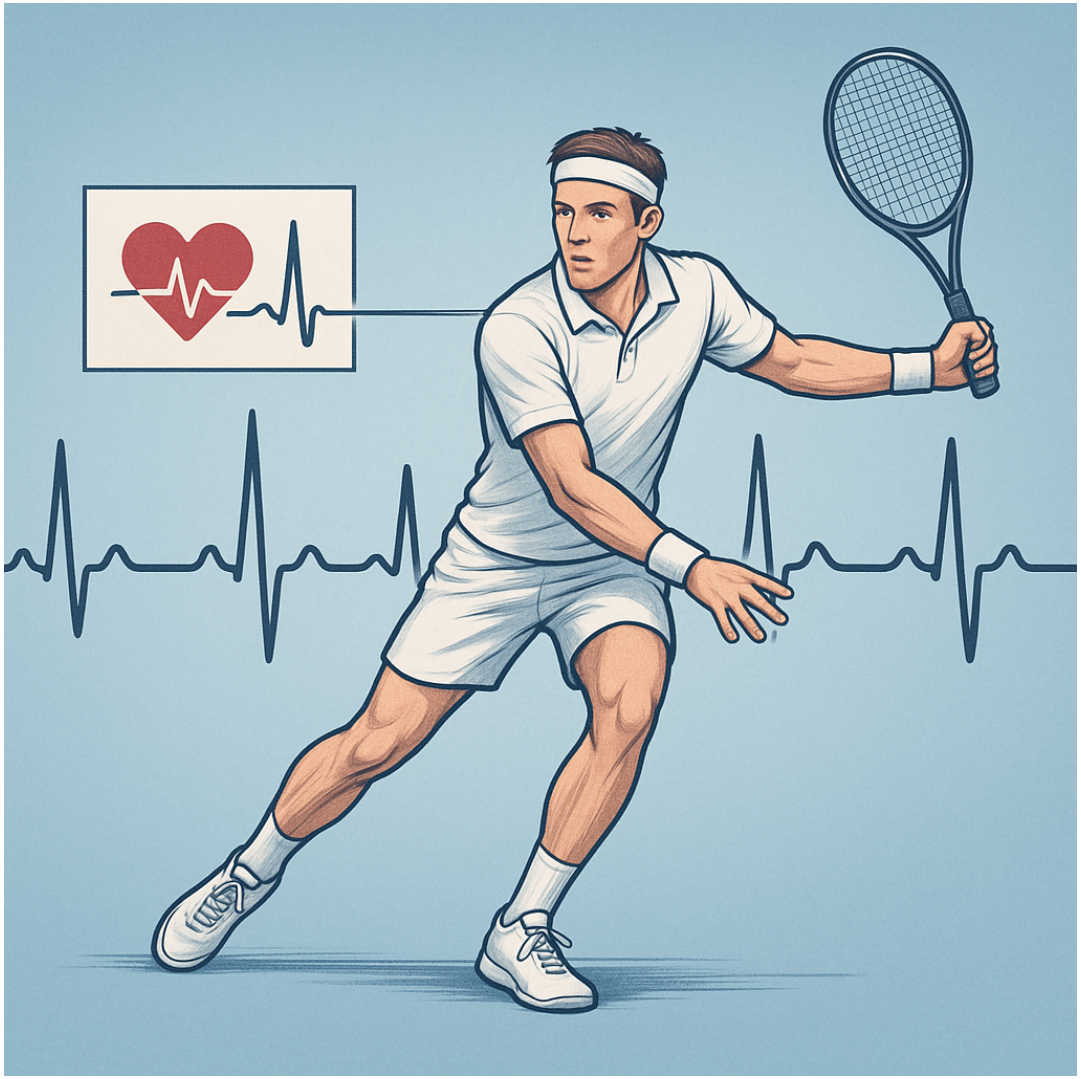


Figure: ECG signal recording during tennis activity

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Solution to Recognizing Motion Artifacts

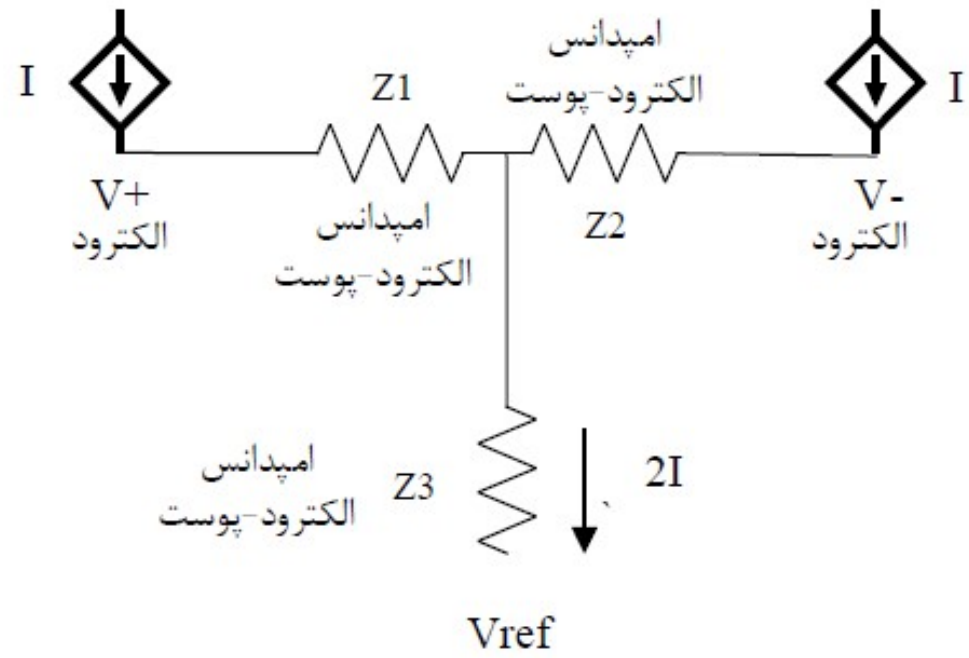


Figure: Injecting two same-direction currents

$$V_{\text{motion artifact}} = (Z_1 + \Delta Z - Z_2) \times I \tag{1}$$

Solution to Recognizing Motion Artifacts

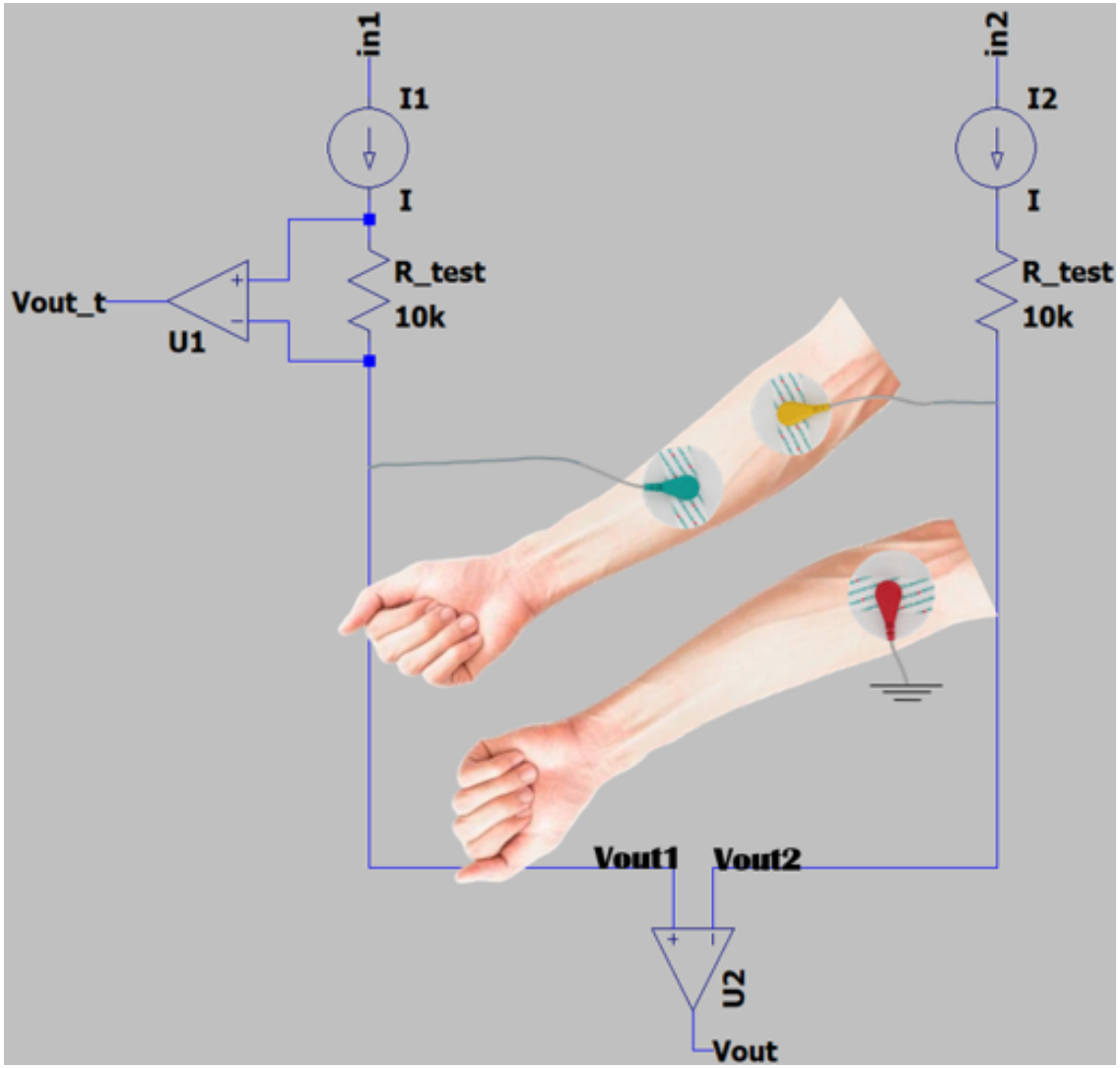


Figure: Application of the current source

Howland Current Source

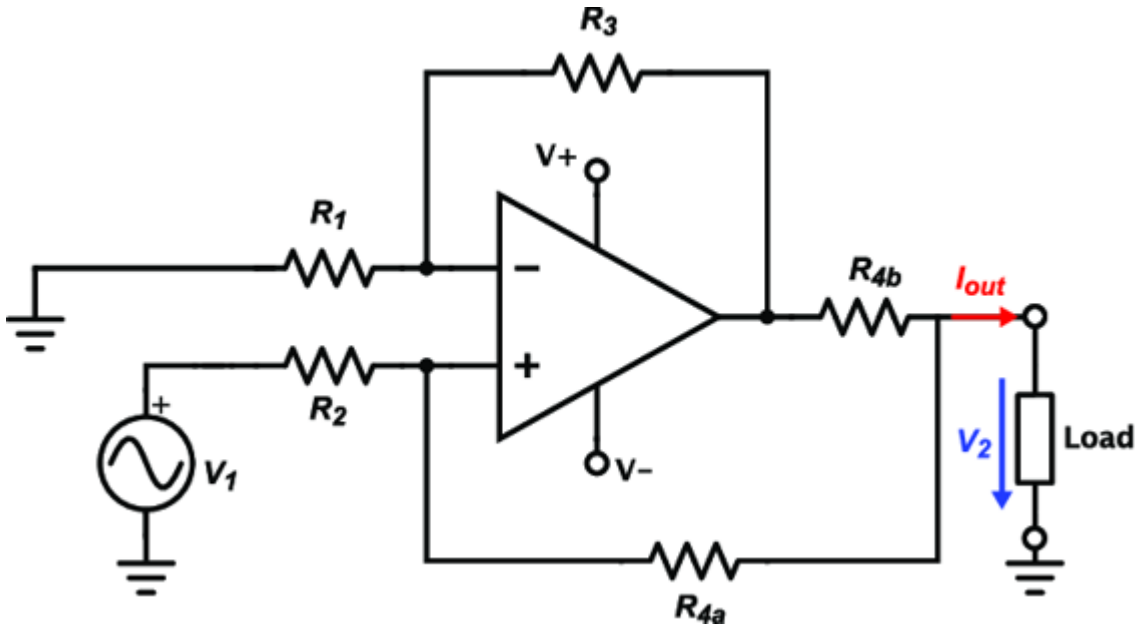


Figure: Howland Current Source.

$$I_{\text{out}} = \frac{R_{4b} + R_{4a}}{R_{4b} \times R_2} V_1 \tag{2}$$

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Project Task Overview

1. Circuit Design
2. Simulation (LTspice, Multisim)
3. Schematic Design (Altium)
4. PCB Design (Altium)
5. Circuit Assembly
6. Circuit Testing

Current Source and Electrode Equivalents

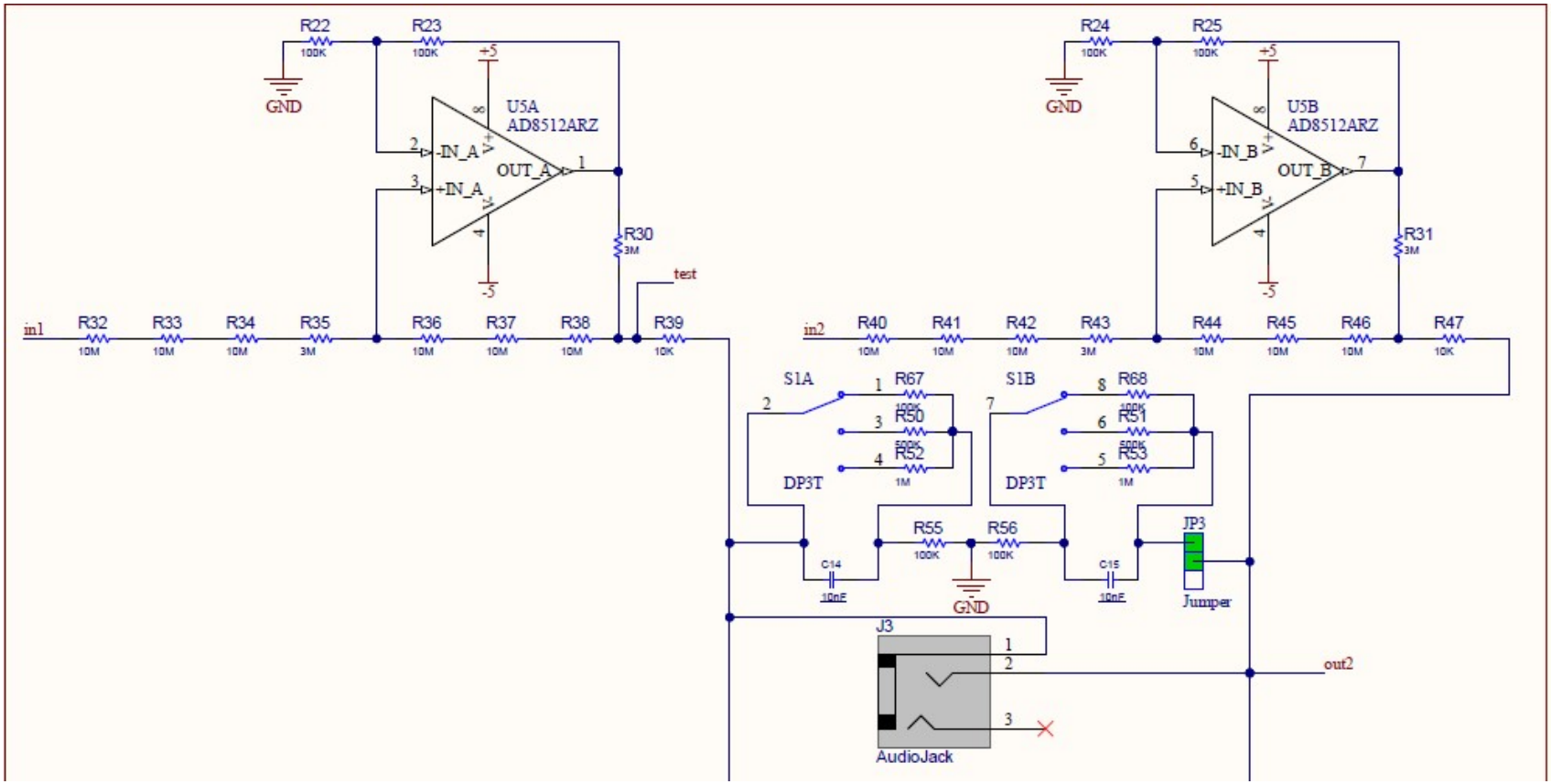


Figure: Equivalent model of the current source and electrodes

Amplifier (AD620)

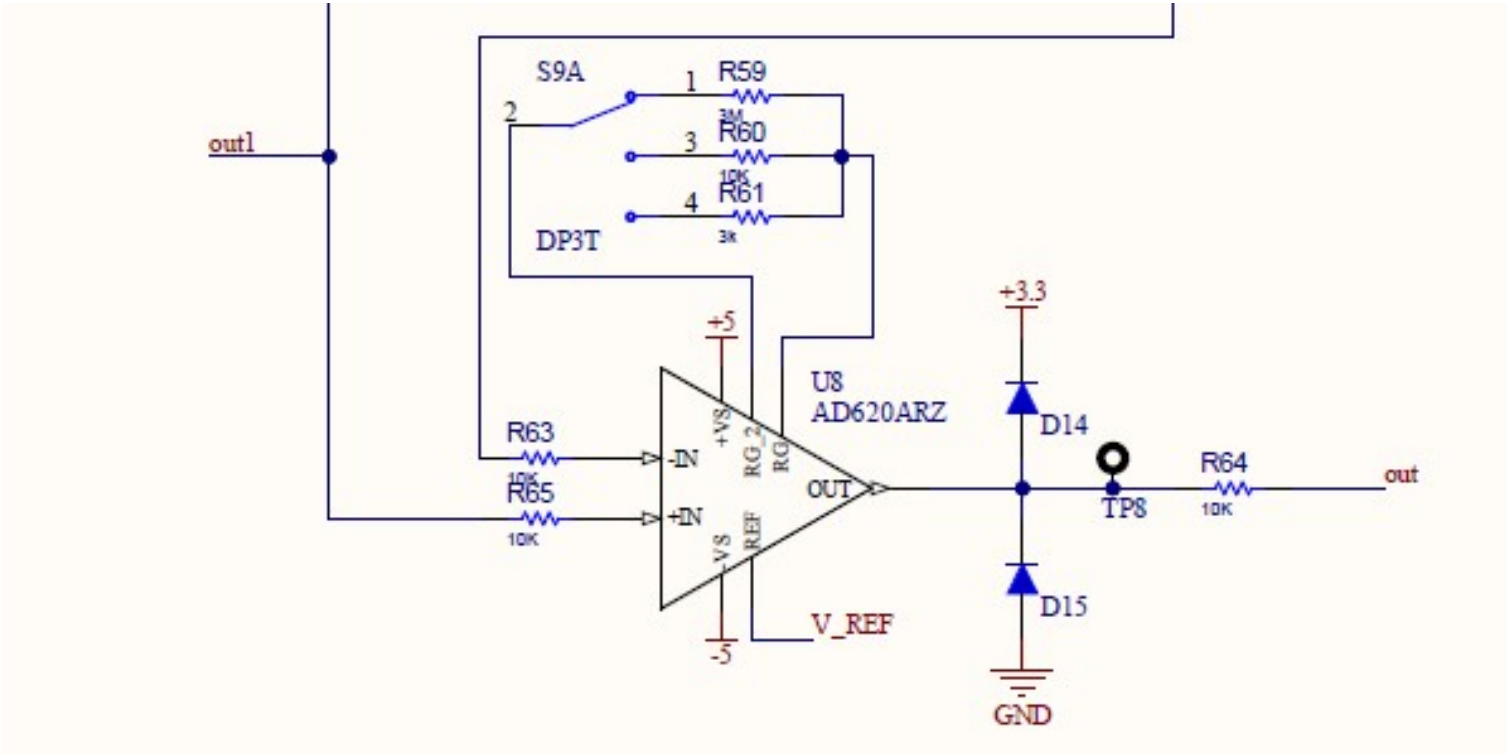


Figure: Amplification stage implemented in the designed circuit

Simulation

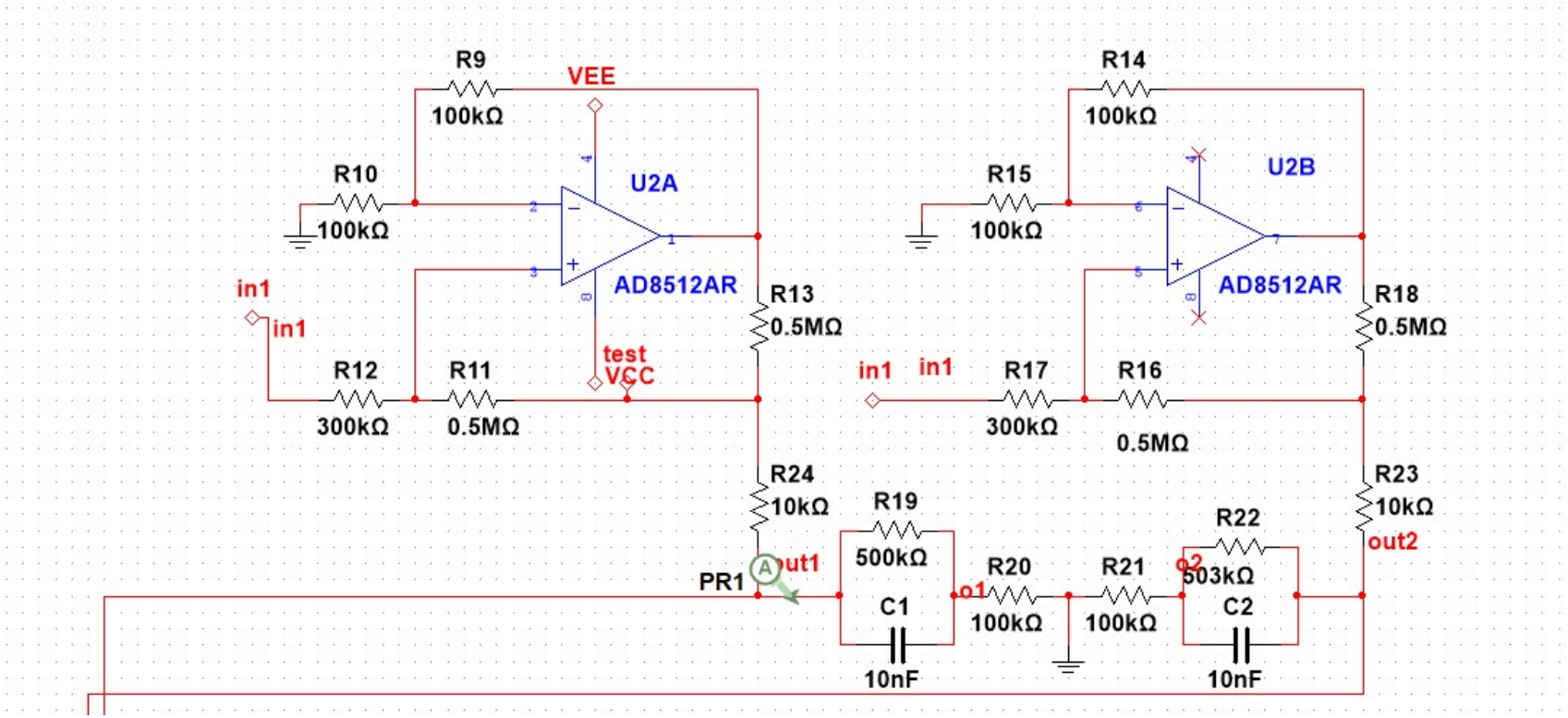


Figure: Circuit Designed in Multisim

Schematic Design

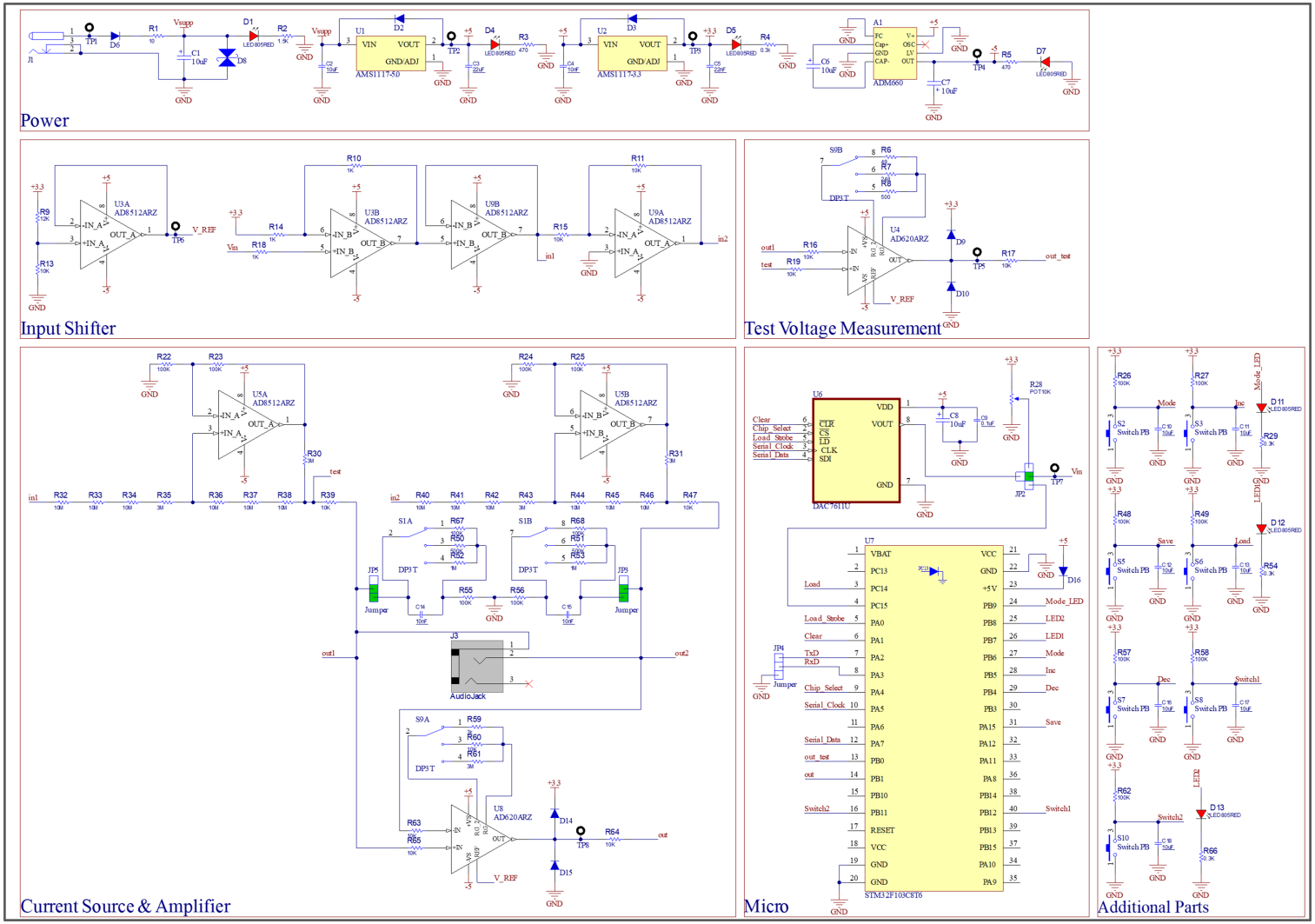


Figure: Schematic of Circuit Developed in Altium Designer

PCB Design

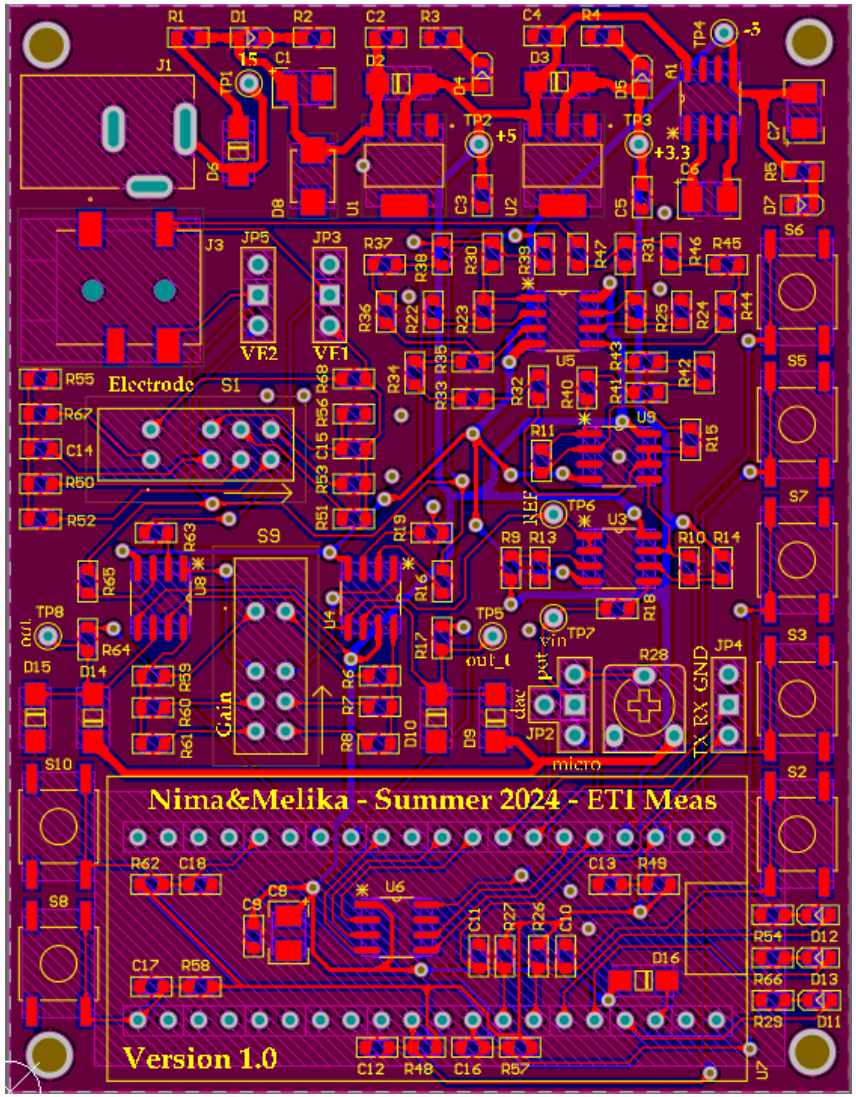


Figure: PCB Outlined in Altium Designer (2D)

PCB Design

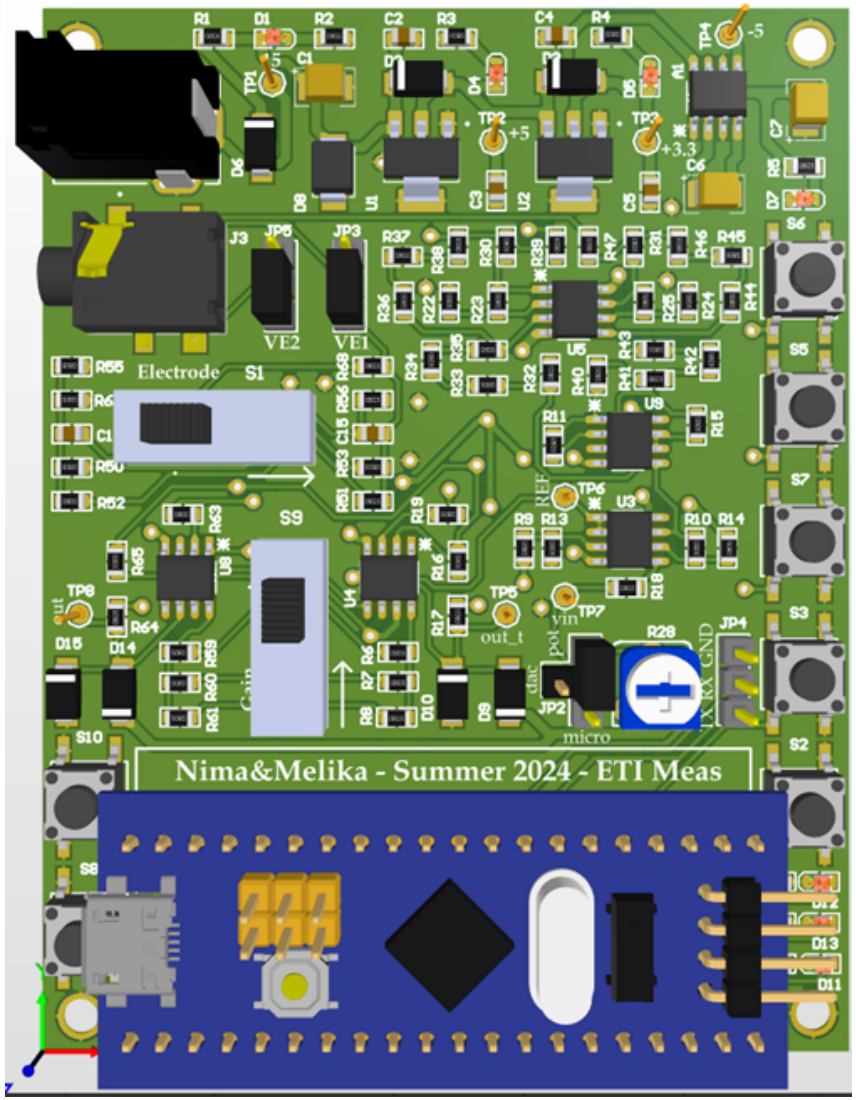


Figure: PCB Outlined in Altium Designer (3D)

Circuit Assembly

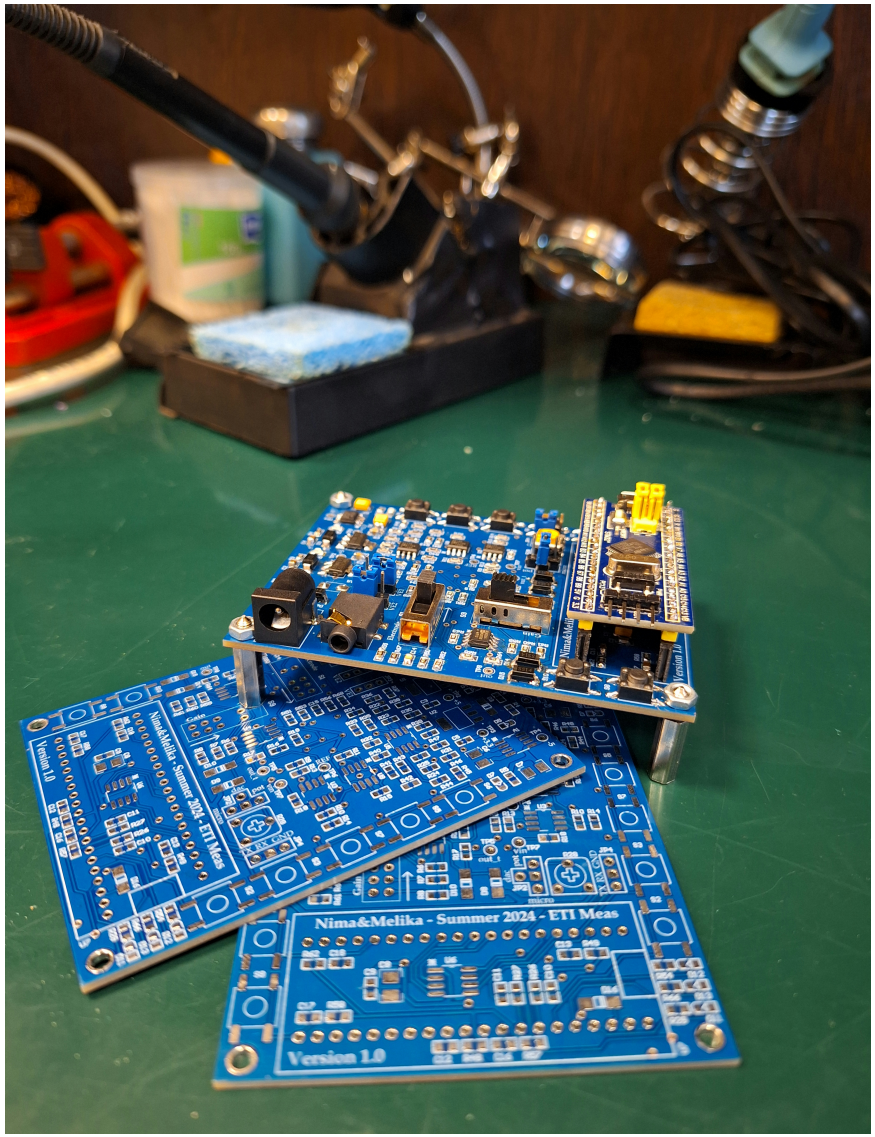


Figure: Assembled Circuit

Circuit Testing

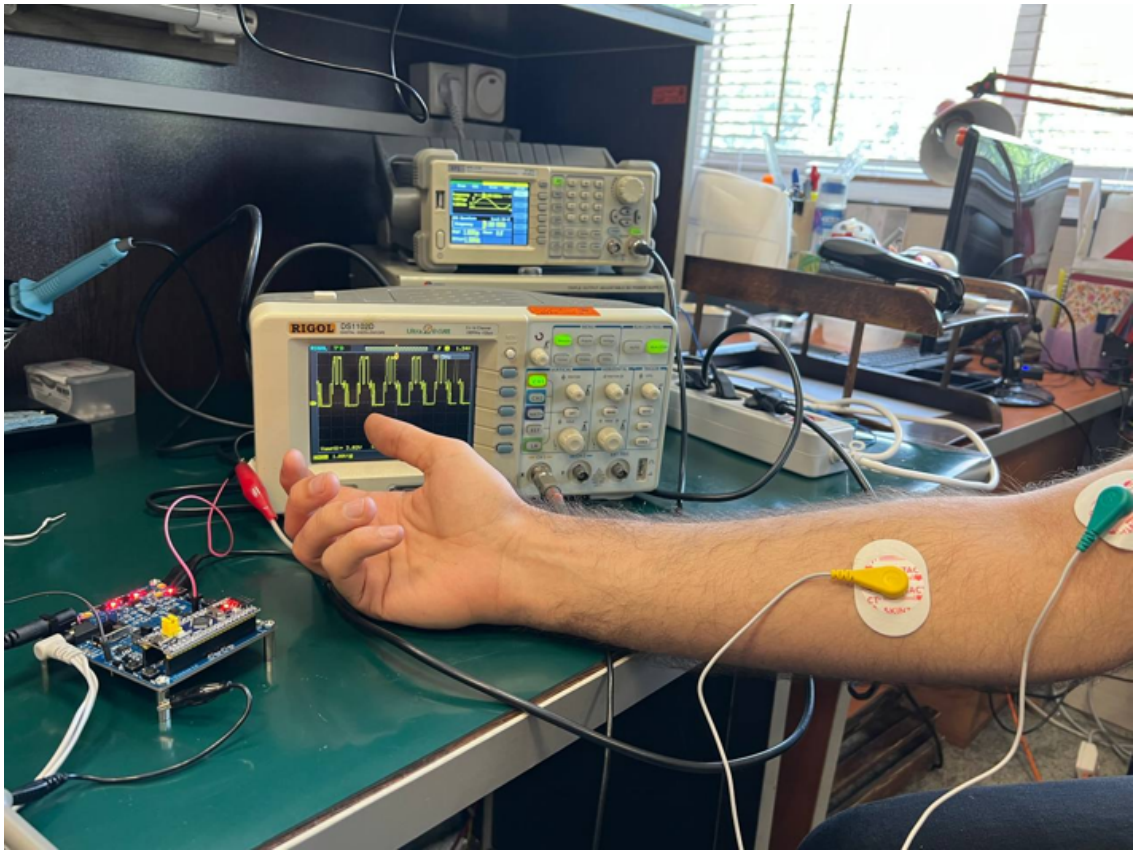


Figure: Measuring Electrode-Skin Voltage

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Signal Processing

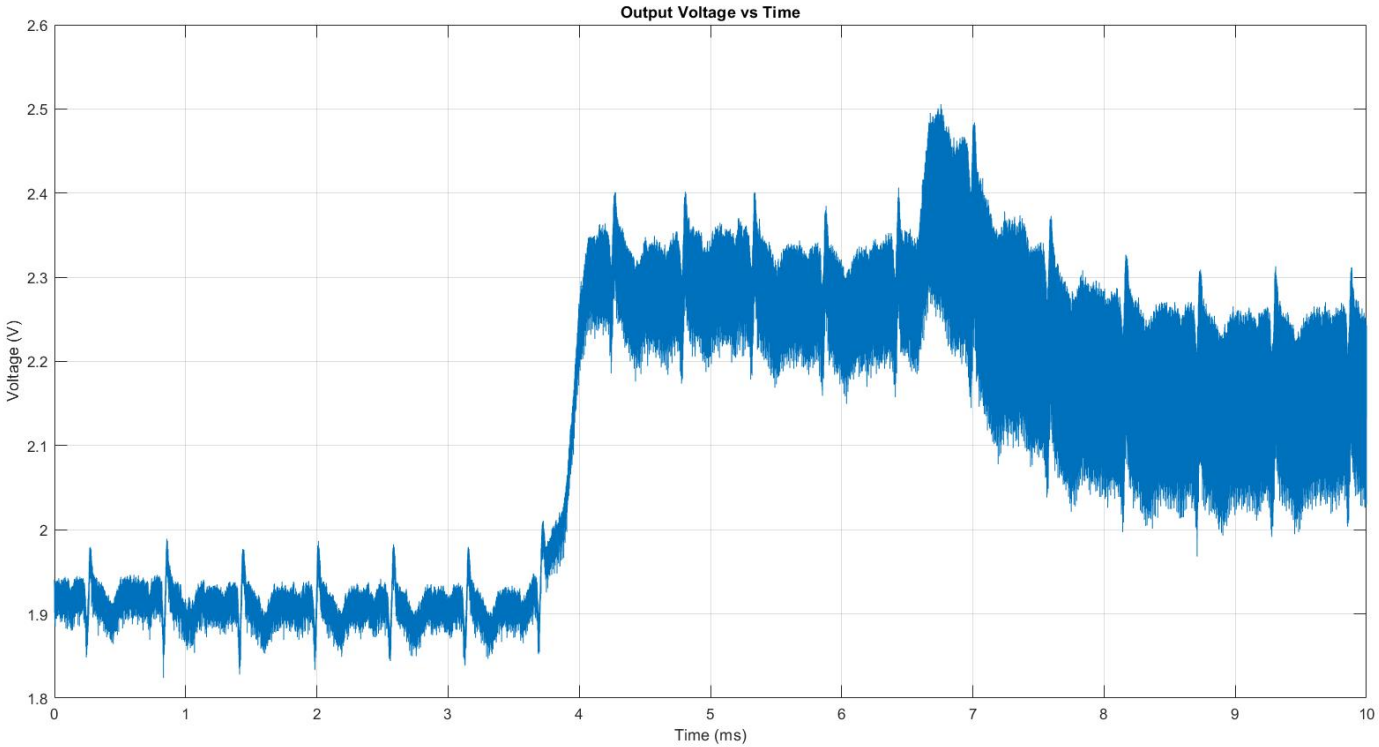


Figure: Circuit output voltage plotted in MATLAB

Output Voltage in the Frequency Domain

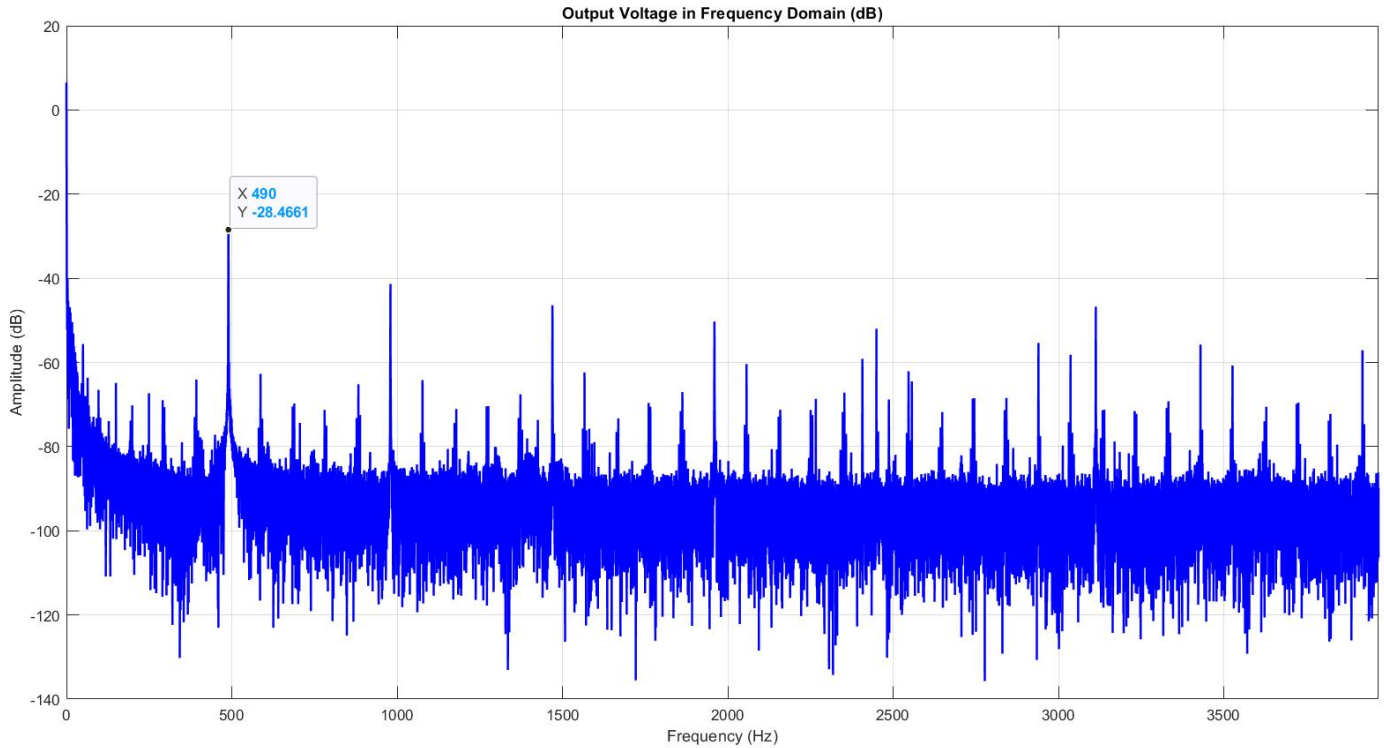


Figure: Output voltage in the frequency domain plotted in MATLAB

Time-Domain and Frequency-Domain Analysis of Filtered Voltage

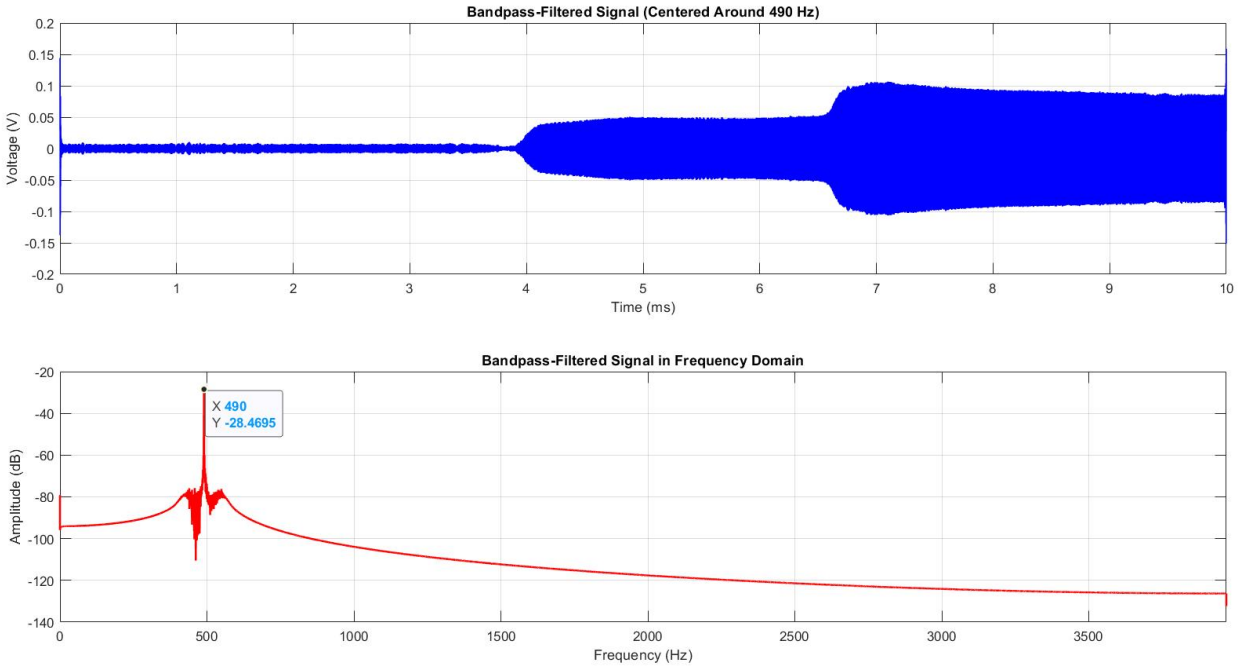


Figure: Filtered output voltage in the time and frequency domains (plotted in MATLAB)

Extracted ECG Signal from Output Voltage

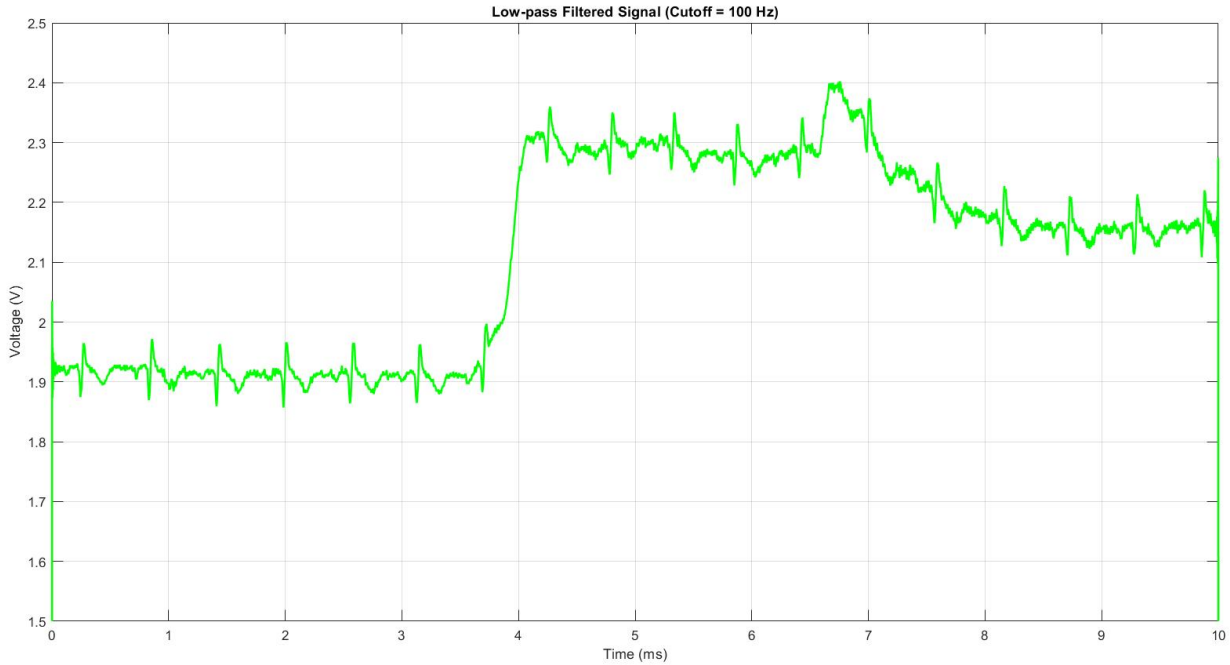


Figure: Extracted ECG signal from output voltage plotted in MATLAB

Envelope of the Filtered Output Signal

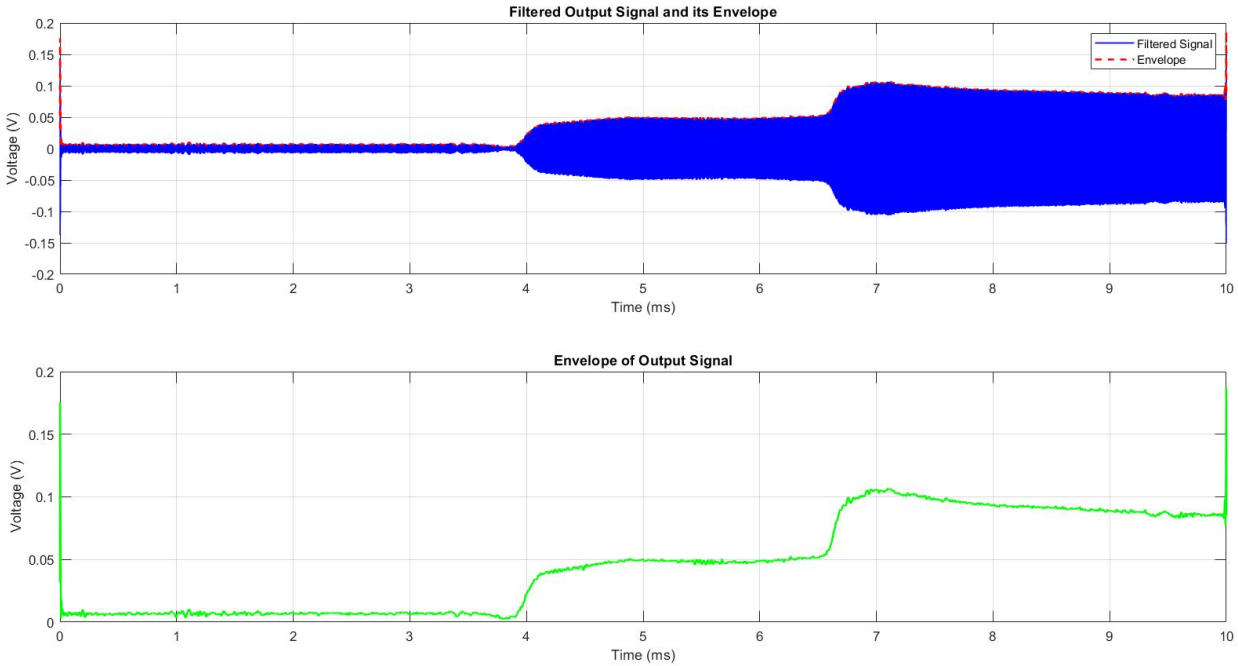


Figure: Time-domain representation of the filtered output signal and its envelope (plotted in MATLAB)

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Results

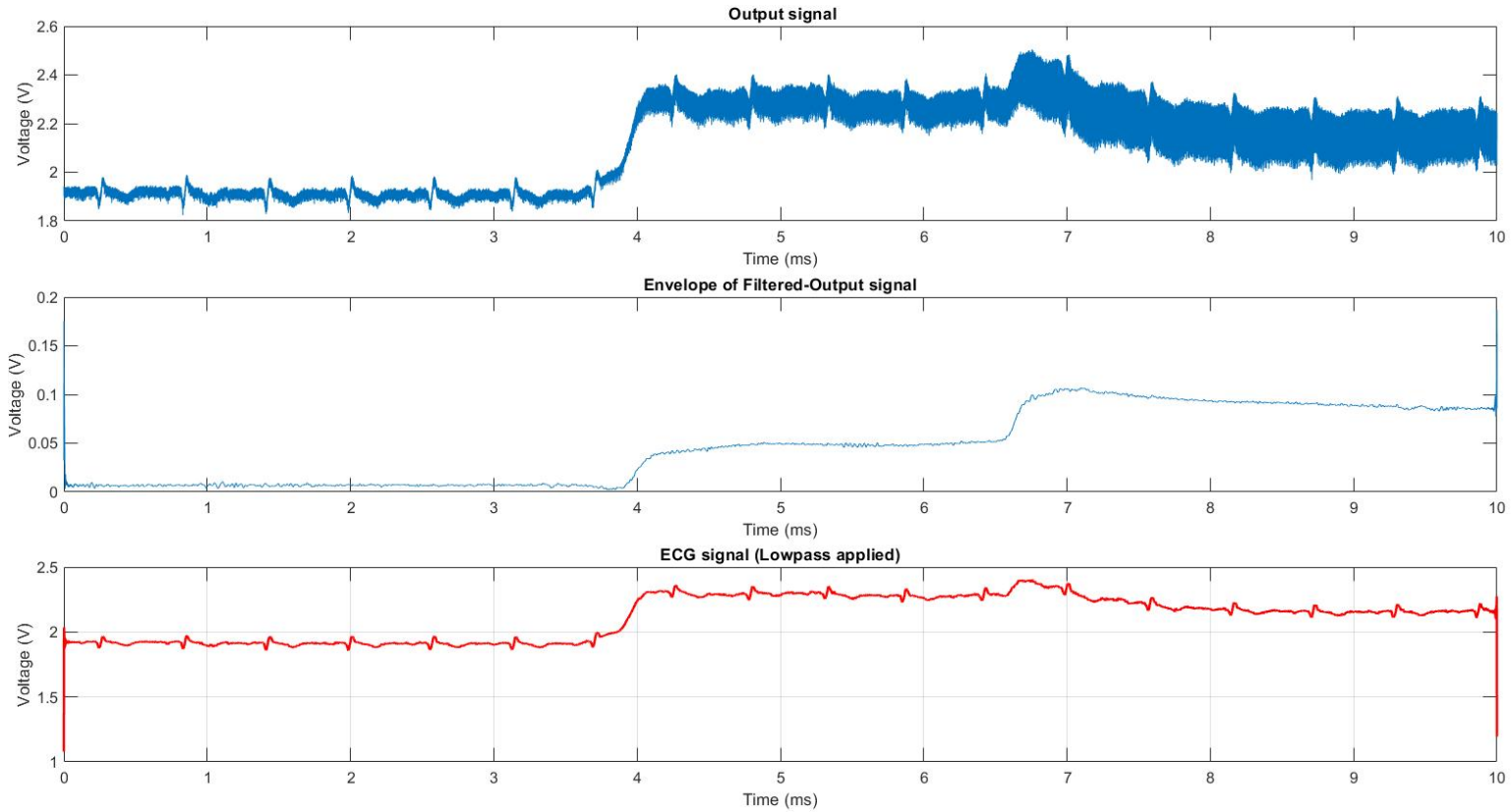


Figure: Time-domain analysis of the ECG extraction process: (top) raw output signal, (middle) envelope of the filtered output, and (bottom) extracted ECG signal after low-pass filtering (plotted in MATLAB).

ETI Signal

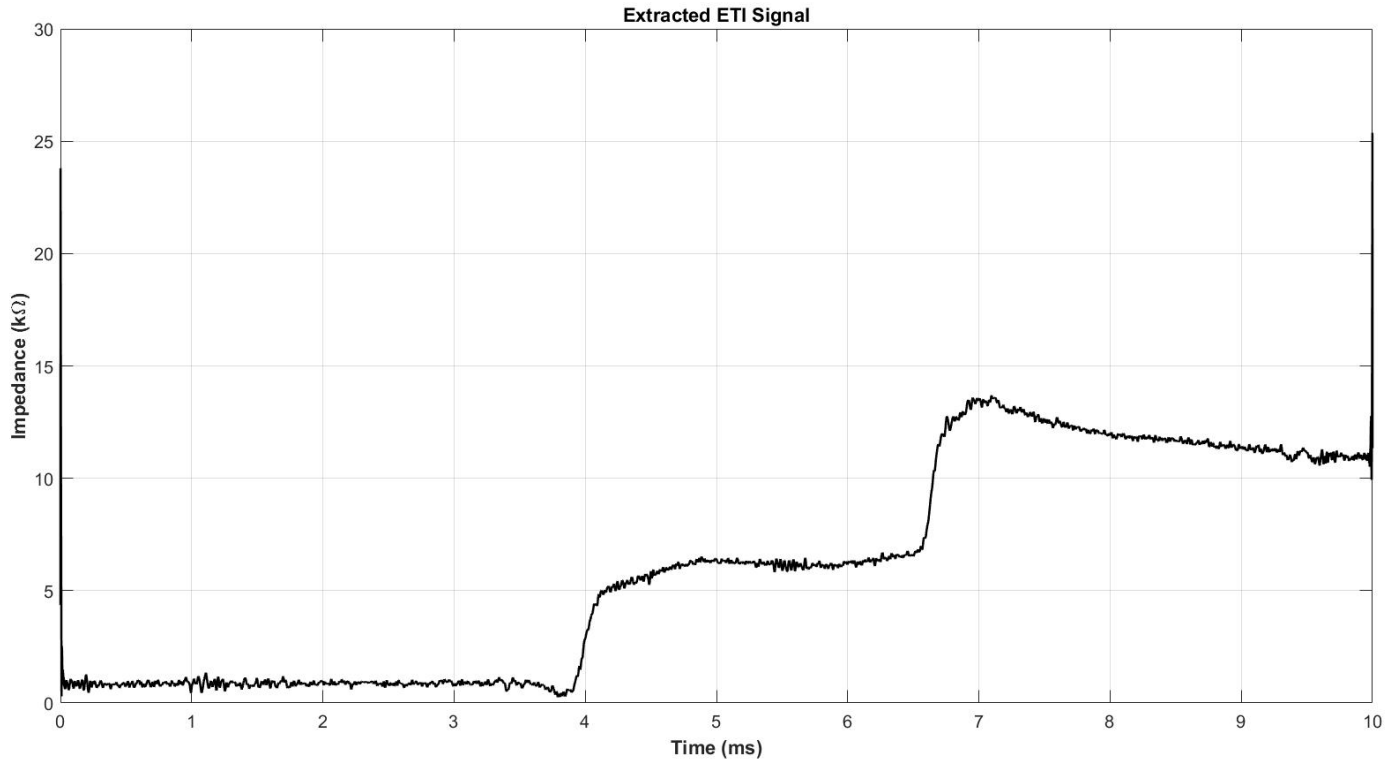


Figure: Extracted ETI signal plotted in MATLAB

Correlation between motion artifact voltage and ECG signal: 70%

Results

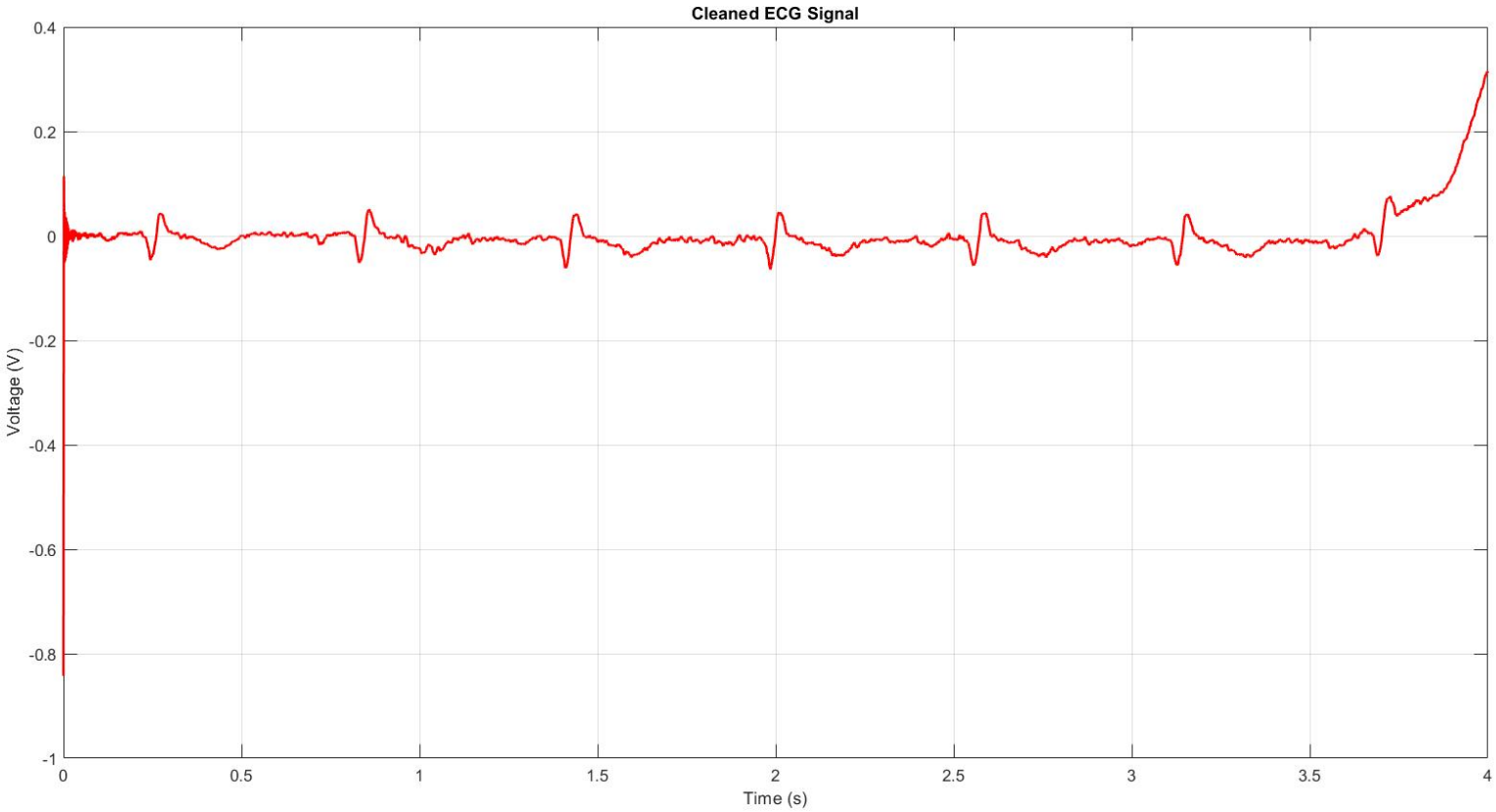


Figure: Extracted and cleaned ECG signal visualized using MATLAB

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Conclusion and Future Works

1. Designing circuit with enhanced capabilities for ECG acquisition
2. Circuit testing planned during physical activity
3. Identifying potential adaptive filters methods for recognizing motion artifacts

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References

1. J. Y. Kim, R. J. Lederman, and D. A. Kass, "Electrocardiographic motion artifact versus electrode impedance," *IEEE Trans. Biomed. Eng.*, vol. 52, no. 1, pp. 136–138, Jan. 2005.
2. T. Seol, S. Lee, and J. Lee, "A wearable electrocardiogram monitoring system robust to motion artifacts," in *Proc. Integrated Nano-Systems Lab.*, Daegu, South Korea.
3. S. R. Wiese, P. Anheier, R. D. Connemara, A. T. Mollner, T. F. Neils, J. A. Kahn, and J. G. Webster, "Correlation between electrode-tissue impedance and motion artifact in biopotential recordings," *IEEE Sensors J.*, vol. 12, no. 12, pp. 3373–3381, Dec. 2012.
4. T. Degen and H. Jäckel, "Continuous monitoring of electrode–skin impedance mismatch during bioelectric recordings," *IEEE Trans. Biomed. Eng.*, vol. 55, no. 6, pp. 1711–1713, Jun. 2008.

Thank you for your attention!

Feel free to ask any questions!